UNIVERSITI TEKNOLOGI MARA
TEST

COURSE	**:**	**FUNDAMENTALS OF ALGORITHMS & COMPUTER PROBLEM SOLVING**
COURSE CODE	**:**	**CSC126**
EXAMINATION	**:**	**JANUARY 2023**
TIME	**:**	**2 HOURS**

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of three (3) parts: PART A (10 Questions)
   PART B (5 Questions)
   PART C (1 Questions)

2. Answer ALL questions from all three (3) parts:

   i) Answer PART A in the Objective Answer Sheet
   ii) Answer PART B and C in your own A4-sized papers. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Objective Answer Sheet – provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 10 printed pages

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PART A

1. Study the code segment below, then answer question that follow:

```
if (item_code == 'A')
      item_price = 59.99;
```

State the purpose of symbol "==" in the `if` condition.

A. It checks if the values of two operands are equal or not, if values are not equal then condition becomes true.
B. It checks if the values of two operands are equal or not, if yes then condition becomes true.
C. It checks if the value of left operand is greater than the value of right operand, if yes then condition becomes true.
D. It checks if the value of left operand is less than the value of right operand, if yes then condition becomes true.

2. Identify statements that are **TRUE**.

I. A relational operator is used to check the relationship between two operands. If the relation is true, it returns 1 whereas if the relation is false, it returns 0.
II. Associativity is only used when there are two or more operators of same precedence**.** Associativity can be either Left to Right or Right to Left.
III. If there are multiple operators in a single expression, the operations are not evaluated simultaneously. Rather, operators with lowest precedence have their operations evaluated first.
IV. Logical operators are used to check whether an expression is true or false. If the expression is true, it returns 1 whereas if the expression is false, it returns 0.

A. I, II and III only.
B. I, II and IV only.
C. II, III and IV only.
D. I, II, III and IV.

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3. Determine which of the following statement is **TRUE** for the code segment given below.

```
if(result >=89) //Line 1
    weight=4; //Line 2

cout<<weight; //Line 3
cout<<result; //Line 4
```

A. If the condition is evaluated as FALSE, the program would jump to execute the statement in Line 4.
B. The given code segment allows to choose between two possible outcomes.
C. Line 3 will only be executed if the condition in Line 1 is evaluated as FALSE.
D. Line 2 will only be executed if the condition in Line 1 is evaluated as TRUE.

4. Code segment given below is an example of _____ loop.

```
int x = 0;
int count =1;
cin >> num;

while (num != x)
{
      sum = sum + num;
      count++;
      cin>>num;
}
cout<<" The sum of" << count<< "is:" <<sum;
```

A. counter-controlled
B. infinite
C. iteration based on data structure
D. sentinel-controlled

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5. Study the given code segment, then choose its FIFTH (5) output:

```
int count =1;
while (count <=30)
{
      cout<< count <<" ";
      count = count + 4;

}
```

A. *9*
B. *17*
C. *21*
D. *29*

6. Change YYYYY to the most suitable statement.

```
for (YYYYY)
{
      cin>> x;
      if (x % 3 == 0)
            cout<< x << "is divisible by 3"<< endl;

}
```

A. `int x =0; x++; x <=10`
B. `x = 1; x <= 10; x--`
C. `int i = 10; x <= 10; i++`
D. `int i = 1; i <= 10; i++`

7. Select statement that is **TRUE**.

A. A value-returning function can return several values.
B. Every user-defined function must return a value.
C. The C++ system allows only user-defined functions.
D. When a function exits, the control goes back to the caller.

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8. Identify the type of these functions:

```
cos(x)
exp(x)
fabs(x)
abs(x)
```

A. Pre-defined function
B. User-defined function
C. Value returning function
D. `void` function

9. Given the function prototype:

```
void guess(int &, int);
```

Choose the INVALID function call statement for the function prototype.

A. `guess(6, 9);`
B. `guess(x, 9);`
C. `guess(x, x);`
D. `guess(x, y);`

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10. Analyze program given below, then predict its output:

```
#include<iostream>
using namespace std;

float funcA(int&, int);

int main()
{
      int a = 50;
      int b = 65;
      int c = funcA(a,b);

      cout<<a<<" "<<b<<" "<<c;
}

float funcA(int &d, int e)
{
    d *= 100;

    return d + e - 2;
}
```

A. *50 65 5000*
B. *50 65 5063*
C. *5000 65 5063*
D. *5000 6500 5063*

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PART B

QUESTION 1

(a) Code segment given below contains FIVE (5) syntax errors. Identify the errors and rewrite the whole code segment with the corrections needed.

```
int main()
{
      int letter;

      cout<<"Enter a letter: ";
      cin<<letter;

      if(letter =='a' && letter == 'A')
            cout <<"To succeed, we must first believe that we can";
      else if(letter == 'e' || letter  = 'E')
            cout <<"Do not ever give up, the beginning is always the
            hardest.";
      else:
            cout<<"\nWrong code entered";

}
```

(5 marks)

(b) Convert the following pseudocode to complete program in C++.

1. Start
2. Read: a number
3. If the number entered is 2 or 6
    a. Write: You have won.
4. If the number entered is 1, 3, 4 or 5:
    a. Write: You are almost right.
5. Otherwise:
    a. Write: You have lost.
6. Stop

(5 marks)

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QUESTION 2

Juara Nursery, a company that sells flowers and plants wants to develop a program to help their staff in calculating the total price to be paid by the customers. The table given below shows the details of flowers and plants available in the nursery:

Category	Item Code	Name	Price per Size (RM)		
			Small (S)	Medium(M)	Large(L)
Decoration (D)	D001	Birah	20	30	45
	D002	Bongsai	70	100	150
Landscape (L)	L001	Pinang Merah	30	60	120
	L002	Pagoda	25	35	45

As a junior programmer at the company, you are asked to write the program. The program will require the user to input the category, item code, size and quantity of flowers or plants bought by the customer. Then, it will calculate and display the total price.

If somehow, the user entered an invalid code for a category or item code, display an appropriate message.

(10 marks)

QUESTION 3

(a) Discuss the difference between pre-test loop and post-test loop in C++.
(b) Give a sample of code segment for both loops to support your answer in (a).

(5 marks)

QUESTION 4

Write a complete program in C++ to calculate the sum for a series of positive number. If the user enters a negative number, the loop will terminate, and the negative number will not be added to the sum.

(5 marks)

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QUESTION 5

A lecturer who is invited as a speaker by any government organization will be paid according to the rate as stated below:

Qualification	Code	Rate Per Hour (RM)
PhD	P or p	500.00
Master's degree	M or m	300.00
Bachelor's degree	B or b	150.00

Write a function definition that accepts the qualification code and number of lecturing hours as its parameters. Calculate the total payment by multiplying the rate per hour with the number of lecturing hours. This function returns the total payment by using `return` statement.

(5 marks)

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PART C

QUESTION 1

Gunung Ledang Resort are doing a promotion in January 2023 for their re-opening session on new year of 2023. Visitors can choose to stay inside Gunung Ledang and experience the natural environment or they can have a single-entry pass for a daily visit.

The daily visit ticket fee is as follows:

Category	Price (RM)
Child (6 – 12) & Senior (60 and above)	2.00
Adult	4.00

The visitor also can choose to stay in Gunung Ledang. There are two options available to spend the night:

Area	Price per night (RM)
Cabin (CB)	50.00
Queen Bed Bedroom (BR)	180.00

Write a complete C++ program that will do the following:

i. Receive input from user either to have a daily visit [1] or stay in [2].
ii. For daily visit, user will have to input the number of adults, children, and senior citizens for the tickets.
iii. For stay in, user can choose choice either to stay in cabin [CB] or in a bedroom [BR].
iv. Calculate the fee for each customer for daily visit or stay in reservation.
v. Print receipt for each reservation.
vi. Calculate the total fee for daily visit and stay in for the day.
vii. Print report to display the total fee for daily visit and stay in.
viii. Format your total fee to TWO (2) decimal places.
ix. The program can take as many visitors as possible.

(15 marks)

END OF QUESTION PAPER

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